

Discussion 2- Sections 2.1 and 2.2

1. A fair coin is tossed four times, and the sequence of heads and tails is observed. Let events A, B, C, and D be given by

A = { at least 3 heads },

B = { at most 2 heads },

C = {heads on the third toss},

D = {1 head and 3 tails}.

If the probability set function assigns $1/16$ to each outcome in the sample space, find

- (i) $P(A)$ (ii) $P(B)$ (iii) $P(C)$ (iv) $P(D)$
(v) $P(A \cap C)$ (vii) $P(B \cap D)$

2. Suppose that $P(A) = 0.40$, $P(B) = 0.50$, $P(A \cup B) = 0.70$. Find

- (a) $P(A \cap B)$ (b) $P(A' \cap B')$ (c) $P(A' \cup B')$.

3. A die is rolled twice. There are 36 possible outcomes. If each of the 36 outcomes has probability $1/36$, assign a value to the probability that the sum of the two dice is

- (i) 3
(ii) between 6 and 8 inclusive.

4. Suppose that $P(A) = 0.60$ and $P(B) = 0.50$.

- (a) Can A and B be mutually exclusive? Why or why not? What is the minimum possible value of $P(A \cap B)$? What is the maximum possible value of $P(A \cap B)$?
(b) What is the minimum possible value of $P(A \cup B)$? What is the maximum possible value of $P(A \cup B)$?

5. Suppose

$$\begin{aligned} P(A) &= 0.40, & P(B) &= 0.34, & P(C) &= 0.55, \\ P(A \cap B) &= 0.19, & P(A \cap C) &= 0.25, & P(B \cap C) &= 0.17, \\ P(A \cap B \cap C) &= 0.07 \end{aligned}$$

- (a) Find $P(A \cup B \cup C)$
(b) Find $P((A \cap B) \cup C)$
(c) Find $P(A \cap (B \cup C))$